Five-Year Medical Expenditure Value of Depression Symptom Resolution and Implications for Outcomes-Based Credits

A US Medical Expenditure Panel Survey-Based Decision-Analytic Valuation

Abstract

IMPORTANCE Current value-based payment arrangements may not recognize medical expenditure differences that accrue after patients achieve depression remission. An outcome-based case rate could make future value visible without evaluating the costs or effectiveness of individual interventions.

OBJECTIVE To estimate the five-year present value of medical expenditures associated with resolution of depressive symptoms and construct a fixed, payer-specific remission credit with alternative recognition levels.

DESIGN, SETTING, AND PARTICIPANTS Decision-analytic valuation integrating nationally representative Medical Expenditure Panel Survey (MEPS) annual expenditure data (2019-2024) with two four-year longitudinal panels (2018-2021 and 2019-2022). The annual models included 77,487 adult person-years, of which 6,604 had a positive depression symptom screen; 4,496 longitudinal participants informed future symptom-state transitions. Analysis was performed in October 2026.

EXPOSURES Current depressive symptom state was classified with the 2-item Patient Health Questionnaire (PHQ-2). A positive-to-negative transition was observed only for the longitudinal screening transition model; the annual expenditure regressions did not follow identified remitters.

MAIN OUTCOMES AND MEASURES Adjusted annual medical expenditure differences, projected five-year expenditure value beginning after the 12-month remission landmark, and fixed case rates after a 53% spontaneous-remission adjustment at 100%, 75%, and 50% recognition. Uncertainty intervals and conditional probabilities that modeled value exceeds each proposed rate were estimated.

RESULTS Adjusted annual symptom-state expenditure differences were $6,746 (95% CI, $5,517-$7,975) overall and $8,499 (95% CI, $6,594-$10,404) for Medicare. The five-year projection yielded 1.302 discounted state-years of differential spending. At 100% recognition, adjusted credits were $4,127 (95% model-based uncertainty interval, $3,335-$4,978) overall; $5,200 ($4,005-$6,467) for Medicare; $1,978 ($1,282-$2,703) for Medicaid; and $1,926 ($1,211-$2,664) for private insurance. The conditional probability that modeled Medicare value exceeded its credit was 50.1%, 98.4%, and greater than 99.9% at 100%, 75%, and 50% recognition, respectively.

CONCLUSIONS AND RELEVANCE The models produced a reproducible provisional symptom-screen rate card, not a directly estimated value of clinical remission. An earlier prospective remitter-specific analysis yielded a discordant and imprecise Medicare estimate. Prospective clinical and claims-based validation is necessary before using these rates to support payment policy.

Key Points

QUESTION
What fixed five-year outcomes-based credit could be assigned to a documented depression remission, without requiring identification of the intervention that produced it?

FINDINGS
In this MEPS-based decision-analytic valuation using 77,487 adult person-years, the estimated Medicare case rate was $5,200 after a literature-based spontaneous-remission adjustment; rates at 75% and 50% recognition were $3,900 and $2,600. Estimated rates were also positive for all-payer, Medicaid, and privately insured populations.

MEANING

The calculation is a reproducible symptom-screen valuation scenario. It is not demonstrated clinical-remission savings: a separate prospective analysis produced a conflicting, imprecise Medicare result.

Introduction

Achieving remission is a central objective of depression care, yet payment arrangements commonly recognize medical spending within a single performance period. This can undervalue improvements that may reduce downstream use over several years. Net Present Value of Care (NPVoC) has been proposed as an approach to recognizing future expenditures associated with a measurable health outcome, rather than limiting incentives to immediate changes in total cost of care.[1]

Maryland's CMMI-approved diabetes outcomes-based credit offers a related precedent: an actuarial valuation of a population-health outcome can be used in financial accountability arrangements even when its benefit extends beyond the current performance year. Depression presents an appealing application because a 12-month remission measure already exists in CMS quality reporting.[2,3,4]

The policy question is deliberately narrow: what fixed dollar amount could be attached to one documented remission, regardless of the intervention used? We developed a payer-specific, five-year medical expenditure rate card using MEPS, longitudinal symptom transitions, and a published estimate of spontaneous remission. The analysis excludes costs incurred before the 12-month remission assessment and does not model intervention effectiveness.[1,2]

Methods

Study Design and Data

We combined pooled 2019-2024 MEPS full-year consolidated expenditure files with two special four-year longitudinal MEPS panels covering 2018-2021 and 2019-2022. The analytic population comprised adults aged 18 to 85 years with usable symptom, survey-weight, demographic, and spending measures. Medicare, Medicaid, and private insurance analyses were estimated in their respective covered populations; the all-medical-expenditure model included eligible adults regardless of payer. Payer domains overlap and are not additive. The study used public-use, deidentified data.[5,6,7,8]

Remission Landmark and Expenditure Valuation

The proposed policy outcome is clinical depression remission at approximately 12 months. MEPS provides the PHQ-2, for which a score of 3 or greater indicates a positive symptom screen and a score below 3 indicates a negative screen. The empirical work did not identify PHQ-9-confirmed clinical remission. Annual expenditure models compared current positive and negative PHQ-2 screens, regardless of previous depression history; the separate four-wave data provided symptom-screen transitions. No group with documented clinical remission and five observed postremission expenditure years was analyzed.[4,9,10]

For each payer, we modeled the association between current PHQ-2 screen status and annual medical expenditures using weighted two-part models: logistic regression for any expenditure and a Gamma log-link model for positive expenditure amounts. The models adjusted for age and age squared, sex, race, poverty, education, insurance, census region, and year. Both predicted screen states were standardized to adults currently PHQ-2-positive in the payer population. Importantly, the screen-negative observations used to fit these models included people without a previously observed positive screen. Estimates were expressed in 2024 dollars; intervention spending was not separately identified.[2,11,12,13,14,15,16,17]

Five-Year Model and Case-Rate Construction

We modeled a hypothetical outcome landmark approximately 12 months after initial positive screening; exact clinical assessment dates were not identified in the expenditure data. Starting after that landmark, one projected trajectory was initialized as screen-negative and the comparison as screen-positive. Four-wave MEPS transition rates, assumed constant over time, determined subsequent screen-state differences. A five-year projection used annual starting-state rewards and 3% real midyear discounting, not five observed years of postremission costs. We applied an external 53% untreated-depression remission estimate as a fixed policy pricing adjustment, then 100%, 75%, or 50% recognition. Neither the 53% discount nor recognition percentages estimated intervention-attributable effects.[18,19,20,21,22]

Statistical Analysis

The expenditure regressions used MEPS self-administered-questionnaire weights; uncertainty was calculated separately using strata- and primary-sampling-unit-aggregated influence functions. The longitudinal transitions used the four-wave SAQ weight and pooled variance-design identifiers. Annual contrasts had normal-approximation 95% survey intervals. Five-year conditional intervals propagated estimated expenditure and transition covariances through 120,000 multivariate-normal simulations, assuming no cross-source covariance and holding the 53% factor fixed. Numerical convergence, selected gradient components,

alternative model specification, and transition arithmetic were checked. No external predictive validation or causal effect estimation was performed.[23,24]

Results

Study Population and Annual Expenditure Differences

The annual analytic sample comprised 77,487 person-years, including 6,604 with positive PHQ-2 screens. The Medicare, Medicaid, and privately insured estimation samples contained 27,468, 15,033, and 45,768 person-years, respectively. Across payer populations, adjusted annual expenditures were higher in the positive-screen state than in the negative-screen state after standardization to adults with current symptoms (Table 1).

Table 1. Adjusted Annual Expenditure Differences Associated With Depressive Symptom State

Payer population	Model, n	PHQ-2+, n	Screen-negative mean, $	Screen-positive mean, $	Difference (95% CI), $
All medical expenditures	77,487	6,604	$9,456	$16,202	$6,746 ($5,517 to $7,975)
Medicare	27,468	2,749	$10,259	$18,758	$8,499 ($6,594 to $10,404)
Medicaid	15,033	2,666	$5,814	$9,047	$3,233 ($2,112 to $4,353)
Private insurance	45,768	2,674	$5,877	$9,024	$3,147 ($1,986 to $4,308)

Note: Predictions were standardized to the covariate distribution of individuals with positive PHQ-2 screens in each payer population; symptom-screen-negative observations in the regression dataset were not restricted to patients with a documented history of depression. Amounts are 2024 US dollars.[9,16]

Five-Year Valuation and Recognition Levels

The four-wave analysis included 4,496 adults. The weighted next-year positive-screen probability was 48.8% among currently screen-positive transition observations and 23.7% among screen-negative observations with any earlier positive screen. Of 444 such previously-positive/screen-negative transition observations, 101 followed another screen-negative wave rather than an immediately preceding positive screen. Their projected difference yielded 1.302 discounted state-years over five subsequent years. With the fixed 53% policy adjustment, the full model-based credit was $5,200 for Medicare and $4,127 for all medical expenditures. These are projected symptom-state valuations, not direct measurements of clinical-remission savings.[18,19]

Table 2. Model-Based Five-Year Case Rates and Conditional Valuation Threshold Probabilities

Payer population	100% recognition, $ (95% UI)	75% recognition, $ (95% UI)	50% recognition, $ (95% UI)
All medical expenditures	$4,127 ($3,335 to $4,978)	$3,096 ($2,502 to $3,734)	$2,064 ($1,668 to $2,489)
Medicare	$5,200 ($4,005 to $6,467)	$3,900 ($3,004 to $4,851)	$2,600 ($2,002 to $3,234)
Medicaid	$1,978 ($1,282 to $2,703)	$1,483 ($961 to $2,027)	$989 ($641 to $1,352)
Private insurance	$1,926 ($1,211 to $2,664)	$1,444 ($908 to $1,998)	$963 ($605 to $1,332)

Table 3. Conditional Probability of Exceeding the Proposed Case Rate

Payer population	100%	75%	50%
All medical expenditures	50.2%	99.5%	>99.9%
Medicare	50.1%	98.4%	>99.9%
Medicaid	50.3%	91.8%	99.8%
Private insurance	50.0%	90.6%	99.6%

Note: UI indicates a model-based uncertainty interval conditional on the 53% spontaneous-remission assumption and specified expenditure/transition models. Threshold probabilities concern modeled valuation uncertainty only; they are not probabilities of Medicare or accountable care organization financial breakeven.[24]

Technical Validation

All four primary two-part models converged. A separately fitted single-part Poisson pseudo-maximum-likelihood model on the same annual observations produced annual contrasts of $7,222 overall, $8,071 for Medicare, $3,345 for Medicaid, and $3,064 for private insurance. A descriptive two-year MEPS record check yielded 46.4% positive-screen persistence, compared with 48.8% in the four-wave analysis; the samples overlapped and this was not independent validation. An earlier prospective remission-specific MEPS analysis, not used to construct the final rate card, instead estimated an imprecise negative Medicare annual contrast of -$7,141 (95% CI, -$17,542 to $3,260). The different results reflect different estimands and remain unresolved (Supplement and audit).[13,25]

Discussion

This decision-analytic model generated positive five-year expenditure valuations by extrapolating adjusted cross-sectional symptom-screen spending differences. These values were not observed in a five-year clinical-remission cohort, and an earlier prospective remission-specific analysis yielded an imprecise Medicare comparison of the opposite sign. Accordingly, the resulting $5,200, $3,900, and $2,600 Medicare amounts are conditional policy-pricing scenarios, not demonstrated savings from clinical remission. A fixed credit remains a possible policy mechanism, but these numerical rates require direct postremission validation.

The policy design separates the clinical outcome from the actuarial price. Once the rate is set, the same credit applies to each qualifying remission, while the recognition percentage lets a payer choose how much of the modeled future value to recognize. This approach is aligned with the NPVoC objective of making future health-related value visible within value-based payment, although it does not by itself measure the savings attributable to an accountable organization.[1,2]

The distinction between valuing remission and financing a program is consequential. The conditional threshold probabilities in Table 2 describe uncertainty in the modeled value relative to a posted rate. They do not reflect intervention effectiveness, care-model costs, or the incremental remissions achieved by a program. A payer therefore should interpret these results as a provisional rate-card valuation rather than a guaranteed return on investment.[3,24]

Limitations

The annual cost comparison was between currently screen-positive and screen-negative adults; many screen-negative observations were not known to have had depression, and model standardization does not identify a remitter-specific causal effect. The four-wave transition model included some screen-negative observations beyond the first year of resolution, and five future years were projected rather than observed. The two-year transition check reused some four-wave respondents, while the independent PPML specification used the same annual observations. The 53% adjustment was held fixed, and the uncertainty intervals omit structural, clinical-measure, and unmeasured-confounding uncertainty. The previously conducted prospective Medicare contrast also had an imprecise opposite sign.[4,9,26,27]

Conclusions

Using observed PHQ-2 screen-state expenditure associations, four-wave screen transitions, and a fixed 53% policy adjustment, we calculated a provisional five-year rate-card scenario: $5,200 for Medicare at 100% recognition, $3,900 at 75%, and $2,600 at 50%. Because a more directly prospective MEPS comparison differs materially, these amounts should not be presented as validated postremission savings or adopted as actuarially established clinical-remission credits without linked clinical and longitudinal claims validation.[1,2,22]

Data and Code Availability

MEPS public-use data are distributed by AHRQ. Reproducibility code, summary results, and diagnostic checks are provided in the accompanying online supplement. Source-input filenames, SHA-256 fingerprints, software versions, and independent validation outputs are documented in the reproducibility archive.[5,8]

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